

Cross-Venue Strategy: OKX x Coinbase

kronos



Team Crater

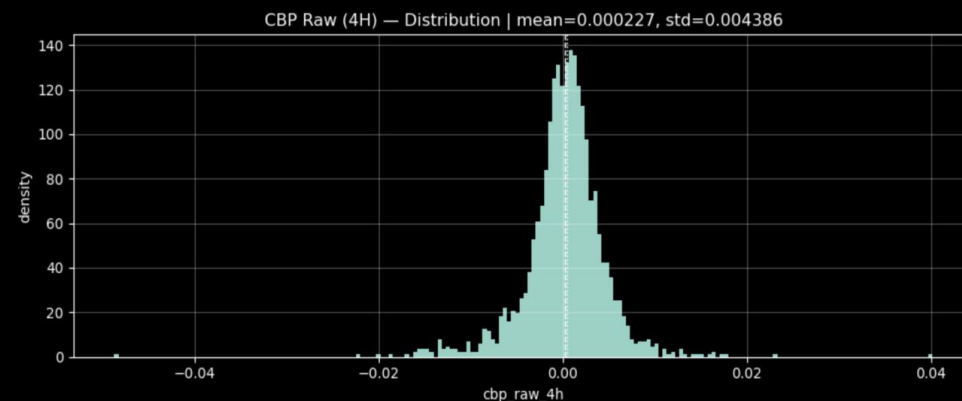
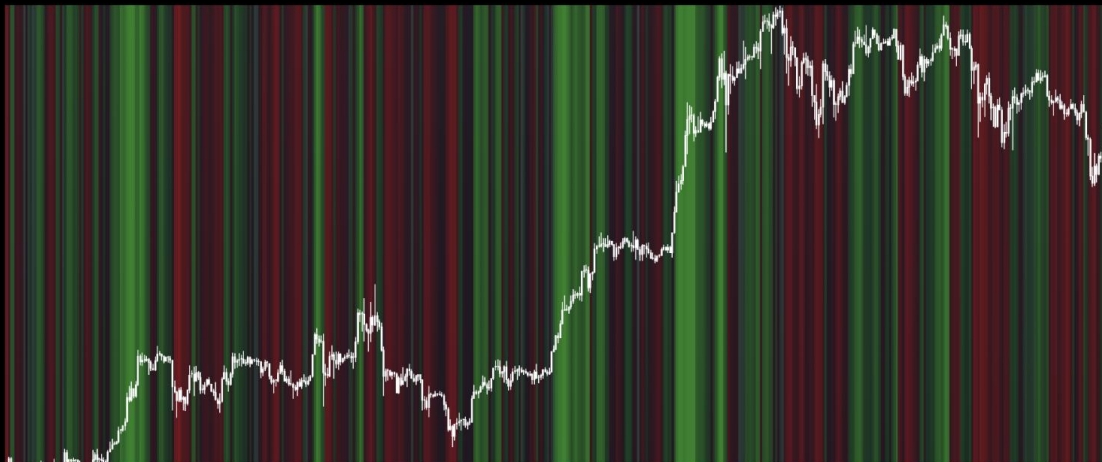
coinbase

2025 NTUFC x Kronos Quant Trading Study Group

Coinbase Premium Alpha

- Coinbase Premium Index (CBP) is the *relative price spread* between a reference “US spot” venue (Coinbase) and our traded venue (OKX perp / OKX price).
- Microstructure interpretation : The premium is a venue-specific imbalance signal, reflecting who is more aggressive (market buys/sells), inventory constraints, and latency/frictions across venues. Coinbase often proxies US-led spot demand; OKX perp often reflects global derivatives positioning.
- “Arbitrage signal” framing for OKX: Think of CBP as a mispricing / basis deviation relative to a reference price. If Coinbase prints higher than OKX, there’s an implied “upward pull” on OKX and vice versa.

OKX close: P_t^{OKX} ; Coinbase close: P_t^{CB} $CBP_t^{\%} = \frac{P_t^{CB} - P_t^{OKX}}{P_t^{OKX}} = \frac{P_t^{CB}}{P_t^{OKX}} - 1$



CBP_Excess Signal

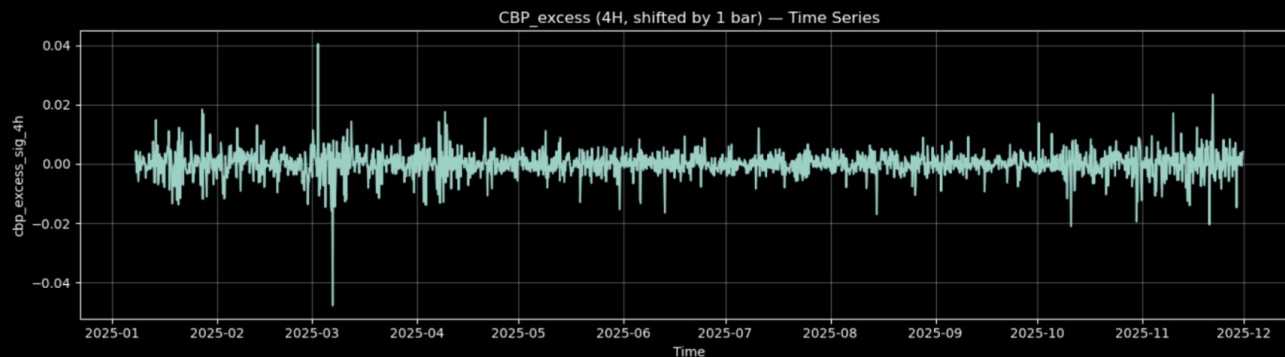
- Baseline premium via rolling mean (CBP_MA window): This estimates the slow-moving structural basis between venues.
- Excess premium (demeaned / high-pass component): temporary dislocation from the typical Coinbase–OKX basis.
 - a. filters out persistent venue bias (fees, stablecoin liquidity, participant mix).
 - b. Stabilizes thresholds: trading on raw CBP can drift; excess is closer to a “zero-mean process”.
 - c. Targets “shock” component: isolates short-horizon deviations that are more likely to mean-revert.
- Band-pass gate: trade only when CBP_excess hit the anomaly threshold → tau_LOW, tau_HIGH.
 - a. If CBP_excess > 0: Coinbase is *unusually rich* vs OKX relative to recent norm.
 - b. If CBP_excess < 0: OKX is *unusually rich* vs Coinbase relative to recent norm.
- Standardization of CBP_excess (Z-score): Take z_t as the directional trading signal and position sizing logic.

$$\mu_t = \text{SMA}_N(\text{CBP}_t) = \frac{1}{N} \sum_{i=0}^{N-1} \text{CBP}_{t-i}$$

$$\text{cbp}_{\text{excess}} : x_t = \text{CBP}_t - \mu_t$$

$$\tau_{\text{LOW}} \leq |\text{CBP_excess}_t| \leq \tau_{\text{HIGH}}$$

$$w = \text{Z_WINDOW}; z_t = \frac{\text{CBP_excess}_t - \mu_t^{(w)}}{\sigma_t^{(w)} + \epsilon}$$

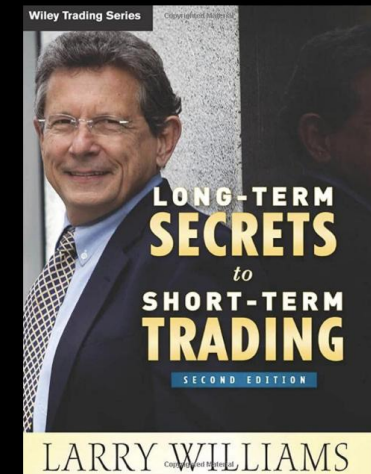
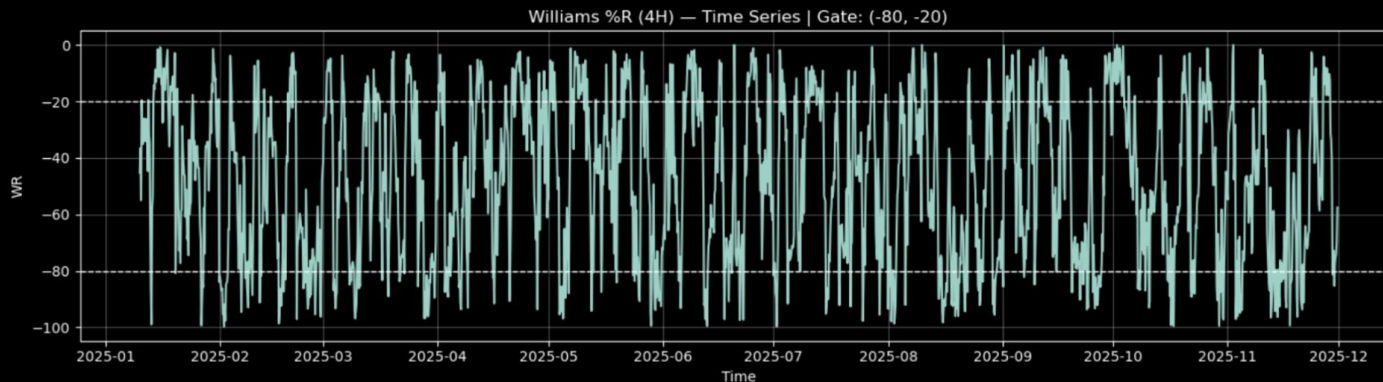


William %R Filter

- Williams %R (Larry Williams), invented by ex Robin cup trading champion, is a bounded oscillator built to measure where the close sits inside the recent high–low range.
- False-signal suppression: A large CB premium move can occur in chop (microstructure noise, temporary dislocations). WR reduces exposure when price is mid-range and mean-reversion dominates. → CBP / event signal can fire during both **real moves** and **random noise**. William %R helps you only take the signal when the market's price action supports it.
- **Confirms momentum**: If we want to go **long**, we prefer price already **closing near the top** of the range (buyers actually in control) and vice versa.
- Reduces overtrading + fees

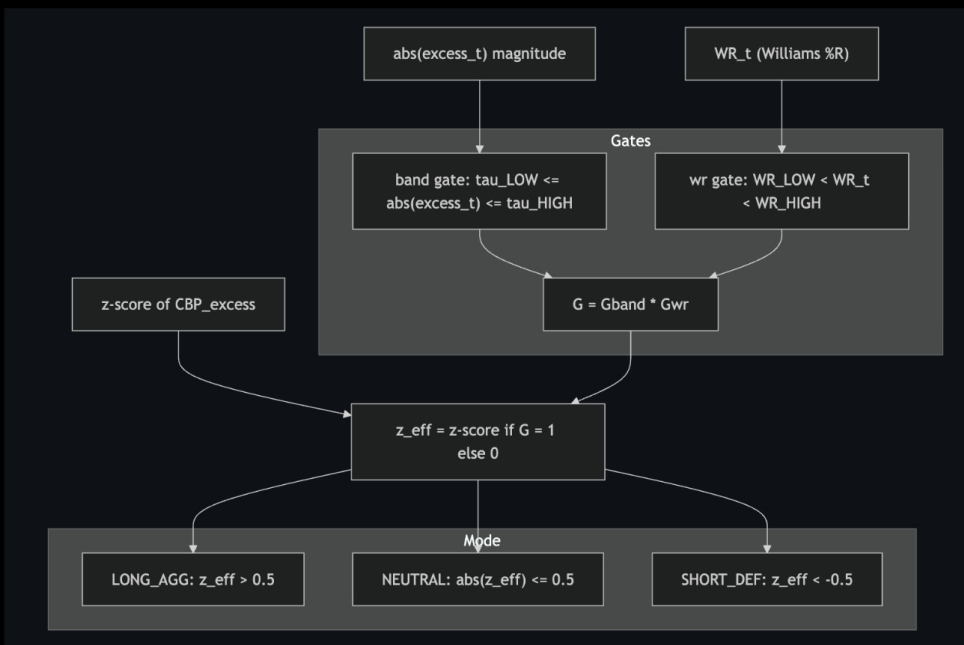
Rolling high/low: $HH_t = \max\{H_{t-n+1}, \dots, H_t\}$, $LL_t = \min\{L_{t-n+1}, \dots, L_t\}$

$$WR_t = -100 \cdot \frac{HH_t - P_t^{OKX}}{HH_t - LL_t + \varepsilon} \quad \text{Gate}_t^{WR} = 1 (WR_{lo} < WR_t < WR_{hi}) = 1 (-80 < WR_t < -20)$$



Trading Algorithms

- Gate signal logic (CBP band-pass + WR Filter)
- If the market is not in your desired regime, the strategy is forced into “no-alpha” mode. $\rightarrow G(t) = 0$



$$GATE_{band}(t) = 1 \left(\tau_{LOW} \leq |x_t| \leq \tau_{HIGH} \right)$$

$$GATE_{wr}(t) = 1 \left(WR_{LOW} < WR_t < WR_{HIGH} \right)$$

$$G(t) = G_{band}(t) \cdot G_{wr}(t)$$

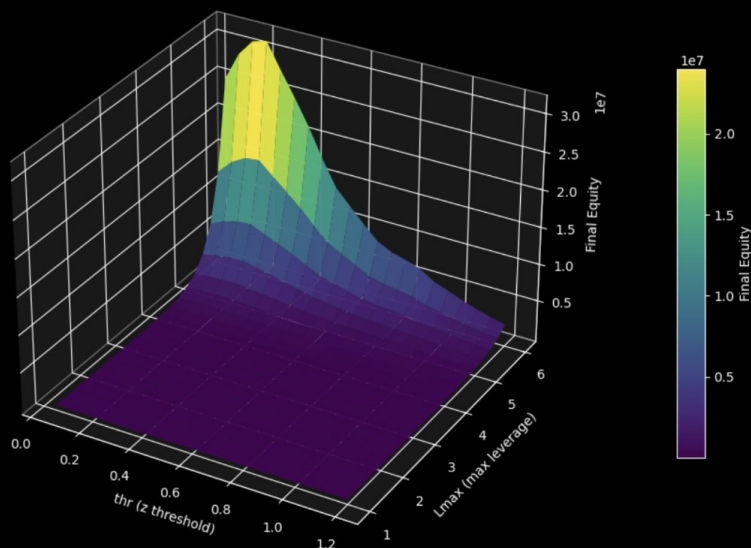
$$z_t^{eff} = \begin{cases} z_t, & G(t) = 1 \\ 0, & G(t) = 0 \end{cases}$$

$$mode(z_t^{eff}) = \begin{cases} LONG_AGGRESSIVE, & z_t^{eff} > 0.5 \\ SHORT_DEFENSIVE, & z_t^{eff} < -0.5 \\ NEUTRAL, & |z_t^{eff}| \leq 0.5 \end{cases}$$

Position Sizing of Alpha

- The formula creates a smooth, non-linear mapping from signal strength to position size. It prevents over-reacting to minor fluctuations (because of the threshold), scales up quickly for meaningful signals (tuned by alpha), and caps exposure at a maximum (via tanh).
- $u_t = \max(z_{thr}, 0)$: This ensures that only signals exceeding the noise threshold contribute to scaling up the position; small signals stay at the baseline.
- Hyperbolic Tan : Tanh smoothly maps any non-negative input into the range $[0, 1)$ and flattens out as the input grows large.

Parameter Surface (Final Equity of BTCUSDT PERP) — thr × Lmax



$$\text{Size}(z) = \text{floor} + (1 - \text{floor}) \tanh(\alpha \max(|z_{eff}| - \text{thr}, 0)).$$

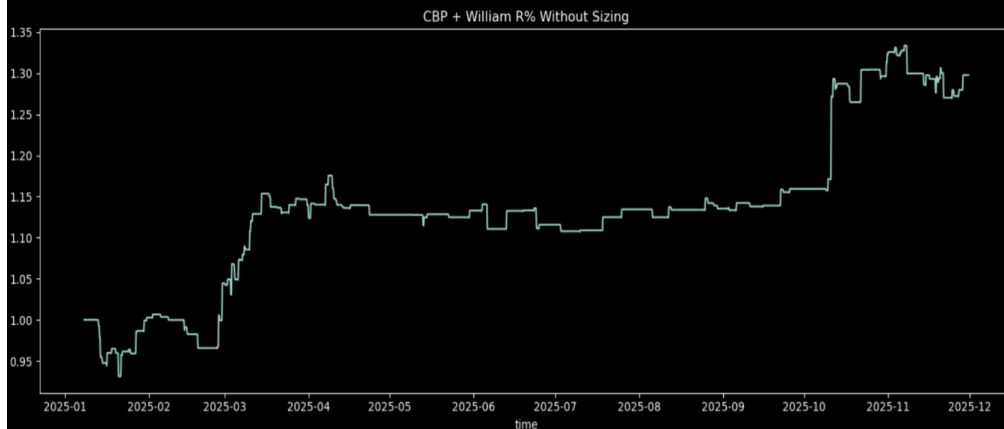
$$u_t = \max(|z_t^{eff}| - \text{thr}, 0).$$

$$\text{Size}(z) = \text{floor} + (1 - \text{floor}) \tanh(\alpha u_t).$$

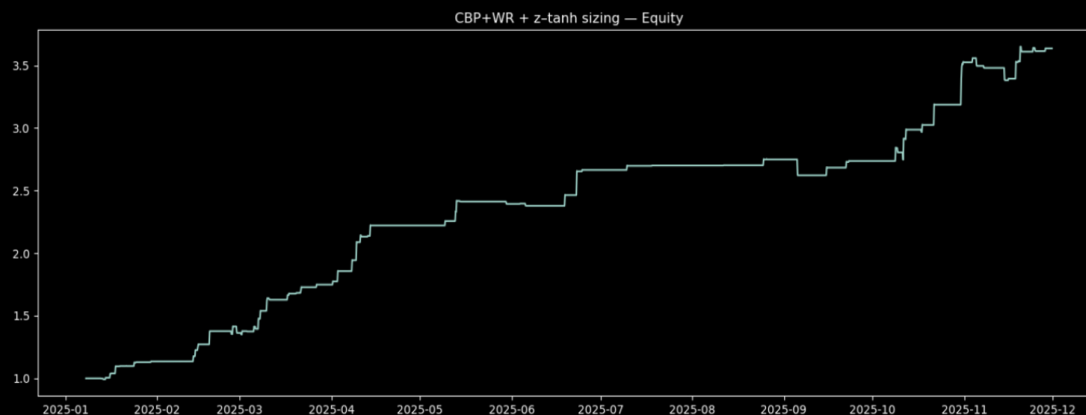
For long entry : $SL = P_{entry} - ATR_t \cdot \text{sl_atr_n}$; $TP = P_{entry} + ATR_t \cdot \text{tp_atr_n}$.

For short entry : $SL = P_{entry} + ATR_t \cdot \text{sl_atr_n}$; $TP = P_{entry} - ATR_t \cdot \text{tp_atr_n}$.

Backtest Metrics

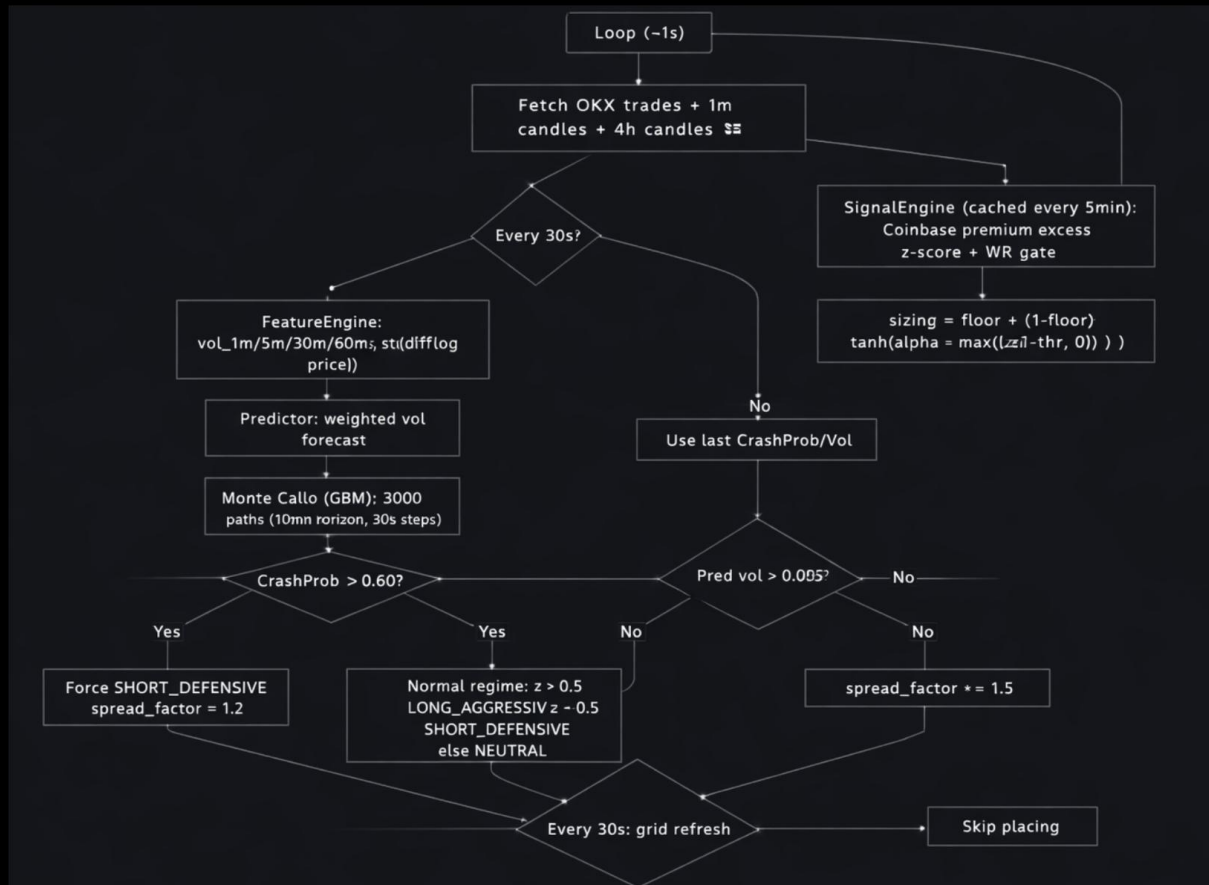


Without Position Sizing
Final equity: 1.336337
Annualized Return : 37.2%
Sharpe (ann): 4.831
Sortino (ann): 15.971
Jensen alpha (ann): 0.325
Beta vs benchmark: 0.007
MDD: -1.311%



With Position Sizing
Final equity: 3.635665
Annualized Return : 310%
Sharpe (ann): 5.061
Sortino (ann): 17.543
Jensen alpha (ann): 1.445
Beta vs benchmark: 0.033
MDD: -4.982%

HFT Grid Trading Execution (Maker Order)



- Built a 30s identification of BTCUSDT swaps, placing multi-level limit orders.
- Long mode: 5buy, 2sell
- Neutral mode: 5buy, 5sell
- Short mode: 2buy, 5sell

Volatility Prediction

$$r_t = \ln\left(\frac{O_t}{O_{t-\Delta t}}\right), \quad \Delta t = 1\text{sec.}$$

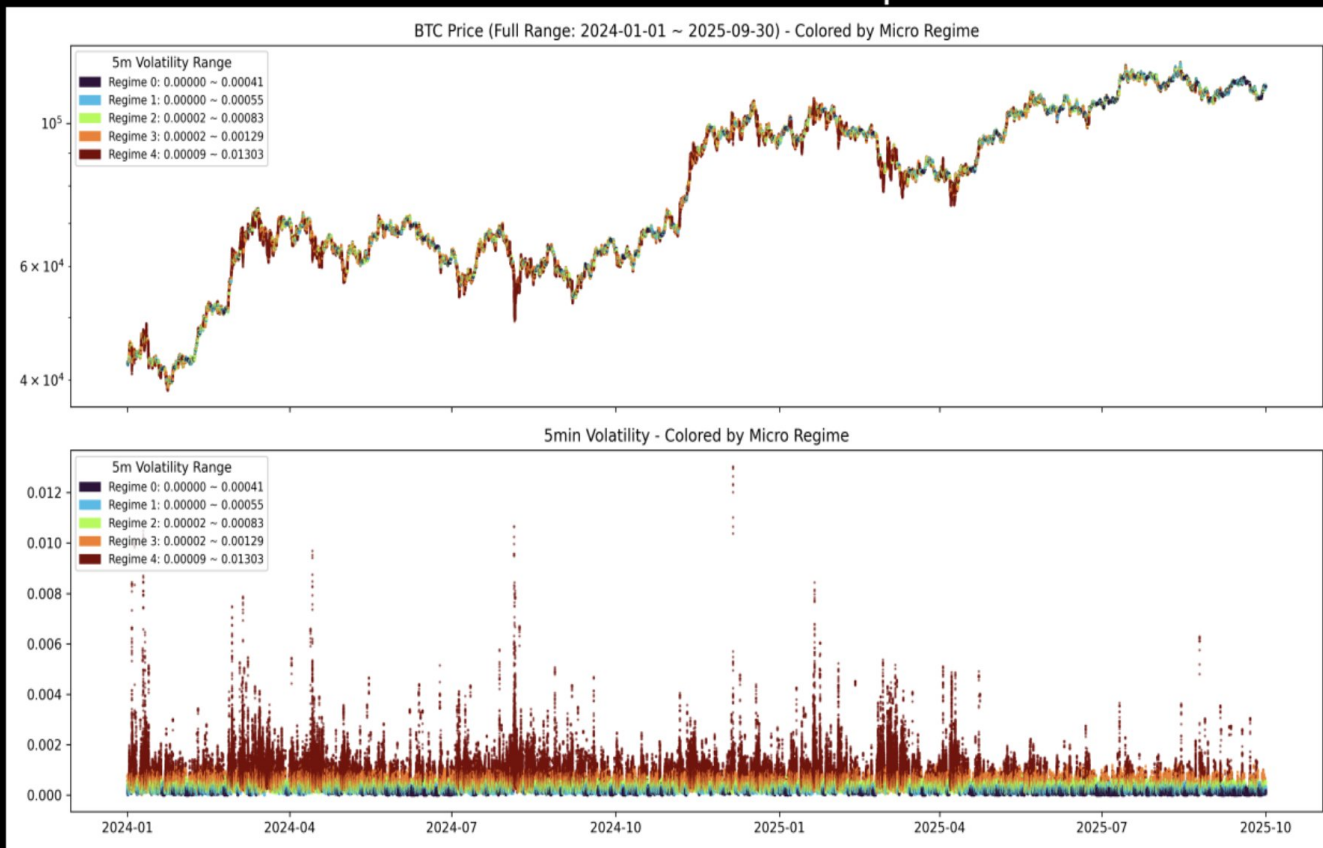
$$RV_t^{(m)} = \sqrt{\frac{1}{m} \sum_{i=0}^{m-1} r_{t-i}^2}$$

$$\mathbf{x}_t = \left(RV_t^{(60)}, RV_t^{(300)}, RV_t^{(1800)}, RV_t^{(3600)} \right)$$

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t)\Delta t + \xi\sqrt{\Delta t}Z_t, \quad Z_t \sim \mathcal{N}(0, 1)$$

$$RV_{t+\Delta t}^{(m)} = \exp(v_{t+\Delta t})$$

$$\widehat{RV}^{mean} = \frac{1}{N} \sum_{n=1}^N RV^{(n)}$$



Grid Trading Algorithm

- Quoted a multi-level limit-order grid around the midprice, placing bids below and asks above at evenly spaced percentage intervals. The spacing automatically widens when predicted volatility rises to reduce adverse selection and churn.

$$s = s_0 \cdot \text{spread_factor}, \quad s_0 = \text{BASE_SPREAD} = 0.002$$

Let grid levels $i=1, \dots, L$ with $L=\text{GRID_LEVELS}=5$.

NEUTRAL (symmetric market-making): $p_i^{bid} = P(1 - si); p_i^{ask} = P(1 + si)$

LONG AGGRESSIVE :
 Buy : $p_i^{bid} = P(1 - 0.8si)$
 Sell : $p_i^{ask} = P(1 + si)$

SHORT DEFENSIVE :
 Buy : $p_i^{ask} = P(1 + 1.2si)$
 Sell : $p_{deep}^{bid} = P(1 - 4s)$

88,063.2	0.6000	9.1935
88,063.1	0.0005	8.5935
88,062.9	0.6000	8.5930
88,062.8	0.6002	7.9930
88,062.7	7.3928	7.3928
88,062.7 ↑ 88,065.0		
88,062.6	3.7497	3.7497
88,062.5	0.0002	3.7499
88,062.4	0.0007	3.7506
88,062.2	0.0005	3.7511
88,062.0	0.0007	3.7518